

REVERSIBLE MODEL MERGING FOR LOW-RANK WEIGHTS

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ABSTRACT

Model merging seeks to consolidate multiple fine-tuned models into weights that retain strong performance across their source tasks. Existing methods have shown that this consolidation can approximate individual model performance in some settings, but they leave a critical and increasingly common regime unresolved: models stored as low-rank representations through low-rank adaptation (LoRA) or post-training singular value decomposition (SVD). We show that applying standard merging methods directly to low-rank weights causes severe degradation in the merged model. This failure motivates a decisive shift in the objective of merging. Rather than collapsing all adapters into one set of weights, we construct a compact basis (e.g., an equivalent of holding two or more models) from which each original task-specific model can be recovered by linear combination. This formulation turns merging into the construction of a reconstruction-capable model space, not the production of a single compromised model. The resulting representation can “revert” to each individual model when needed, reflecting the practical reality that a single merged model cannot reliably dominate a model specialized for its own task. We introduce Reversible Model Merging (RMM), an efficient, data-free, and flexible method with a closed-form solution for selecting the optimal basis of model weights and task-specific linear-combination coefficients. Extensive experiments across diverse datasets and model scales demonstrate that RMM consistently and substantially outperforms existing merging approaches, preserving the performance of low-rank compressed models with a clear margin.

1 INTRODUCTION

Parameter-efficient fine-tuning methods, such as Low-Rank Adapters (LoRA) (Hu et al., 2021), are now a central mechanism for adapting large language models to diverse downstream tasks without paying the full cost of retraining. In parallel, model compression techniques such as post-training singular value decomposition (SVD) truncation have become a standard route to reducing storage and inference costs, adding a complementary axis of efficiency (Ryu et al., 2023; Ping et al., 2024; Wang et al., 2025b;a). Model merging has advanced alongside these developments as a direct way to consolidate the knowledge of multiple task-specific models into one deployable model. Methods ranging from parameter interpolation to optimization-based merging show that a merged model can perform competitively across source tasks without access to the original training data (Matena & Raffel, 2022; Yang et al., 2024; Yadav et al., 2023; Wei et al., 2025).

This progress nevertheless leaves a decisive limitation: even sophisticated merging strategies do not match the original individual fine-tuned models. The gap becomes especially large when the models to be merged are already compressed into low-rank form, either through LoRA fine-tuning or post-training SVD truncation. A single merged model can serve multiple tasks, but it does so by collapsing distinct task-specific representations into one parameter space. That collapse creates task interference and removes specialized adaptations that each model learned for its own task (Gargiulo et al., 2025).

Low-rank adapters and SVD-truncated models are compact individually, yet collections of them create a concrete scalability bottleneck. Practical multi-task, federated, and continual learning systems can accumulate dozens or hundreds of task-specific low-rank adapters, each with its own storage and management overhead. Retaining every adapter weakens the storage savings that low-rank pa-

parameterization is meant to deliver and complicates deployment. The right merging strategy must therefore operate directly on low-rank representations, both to reduce redundancy and to support efficient sharing, deployment, and transfer of compressed models across tasks.

We first expose this limitation by evaluating existing merging techniques on low-rank compressed models. Our analysis focuses on deltas, the weight differences between each fine-tuned model and the shared pre-trained initialization. Conventional merging methods combine these deltas into one parameter set and implicitly assume that the resulting directions are linearly compatible. Our experiments show the opposite in the low-rank regime: directly merging low-rank deltas causes catastrophic performance degradation. The failure follows from two reinforcing sources: low-rank representations have limited expressive capacity, and independently optimized task-specific subspaces are misaligned, producing severe interference when forced into a single merge.

This observation motivates a sharper question: merging need not end in a single final model. If we allow a compact representation equivalent to retaining more than one model, we can preserve the ability to reconstruct the original models on demand. We therefore maintain a small set of weights that acts as a shared basis from which each original low-rank adapter can be recovered through linear combination. This reframing opens the central design space of this paper: instead of compressing all information into one model, we preserve complementary representational directions across a small basis and enable post-merging “reversion” to the original adapters. Our approach rejects the assumption that merging must terminate in one model and defines merging as the construction of a reconstruction-capable model space.

Building on this reconstruction-driven perspective, we develop a theoretical framework that identifies the optimal basis for minimizing reconstruction error. The result is Reversible Model Merging (RMM), which recasts merging as basis selection. The objective is explicit: find a compact set of task-vector directions that accurately reconstructs all original models. RMM provides a closed-form solution that jointly selects the optimal basis task vectors and the task-specific coefficients needed to recover each adaptation by linear combination. Across datasets and model configurations, RMM consistently surpasses conventional merging strategies by a substantial margin while preserving the efficiency benefits of low-rank parameterization. The method also exposes a practical control knob: a tunable hyperparameter sets the number of retained reversible components and thereby determines the storage–performance trade-off.

2 RELATED WORK

Model merging is a powerful strategy for consolidating knowledge from multiple fine-tuned models. Existing methods fall into two main categories: data-free parameter-space manipulations and optimization-based approaches that use data or auxiliary objectives. We review representative techniques from both lines and position our work against their shared single-model merging objective:

Task Arithmetic: Task arithmetic is a foundational model-merging technique (Ilharco et al., 2023). It defines a delta as the difference between a fine-tuned model and its pre-trained initialization, then treats these deltas as modular directions in weight space. Adding a delta can impart a capability to the base model, subtracting one can suppress an unwanted behavior, and scaling or combining deltas can blend multiple capabilities. This linear view established the basic weight-space language for composing model behavior.

DARE: Simple delta addition is effective in favorable cases, but it breaks down when parameter updates overlap or conflict. DARE (Drop and Rescale) directly targets this failure mode through aggressive randomized pruning (Yu et al., 2024). It discards most parameter changes—up to 90-99%—under the premise that many updates are nonessential or interfere with other models. The surviving updates are rescaled to preserve the magnitude of the modification, retaining core task-specific changes while reducing interference.

TIES: TIES-merging replaces DARE’s randomized pruning with a deterministic conflict-resolution pipeline (Yadav et al., 2023). Its three stages—Trim, Elect Sign, and Merge—systematically remove low-magnitude updates, enforce sign agreement at each parameter position, and average the remaining sign-consistent parameters. This consensus-driven procedure reduces destructive interference and produces a more stable single merged model. EMR-Merging (Elect, Mask and Rescale) pursues

the same interference-reduction goal by electing a unified model and attaching task-specific binary masks and rescalers that shift the unified model toward each task (Huang et al., 2024).

Other Merging Methods: Beyond parameter-space manipulation, several merging methods introduce optimization or require access to a small dataset. Model soups (Wortsman et al., 2022) tune merging weights for task arithmetic by minimizing validation loss, steering interpolation toward better generalization. Fisher-weighted averaging (Matena & Raffel, 2022) uses second-order information to assign higher weight to parameters estimated to matter more for task performance. RegMean (Jin et al., 2023) and AdaMerging (Yang et al., 2024) explicitly optimize task-dependent merging coefficients using objectives based on prediction discrepancy and unlabeled prediction entropy, respectively. Another recent method minimizes the loss difference between the merged model and the individual fine-tuned models through a first-order Taylor surrogate (Wei et al., 2025). These methods can improve single-model performance, but their reliance on data, additional optimization, or both limits their applicability when merging must remain data-free and retraining-free. More importantly for our setting, they retain the same single-model endpoint that fails for low-rank compressed weights.

3 PRELIMINARIES

We start from the standard model-merging problem: consolidate a collection of fine-tuned models into a single unified model. Let $\{\theta_1, \dots, \theta_n\}$ denote n models fine-tuned on n distinct tasks. A merging algorithm $\mathcal{M}$ maps these models to one weight set $\theta_{\mathcal{M}} = \mathcal{M}(\{\theta_i\}_{i=1}^n)$. The conventional objective is to choose $\mathcal{M}$ to maximize average validation performance across tasks:

$$\max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\theta_{\mathcal{M}}, \mathcal{D}_i), \quad (1)$$

where $\mathcal{D}_i$ is a held-out validation set for task i , and Perf is an appropriate metric such as accuracy or F1 score. As in standard merging, all fine-tuned models are assumed to share the same pretrained initialization θ_{pre} . For any layer l , the update (or delta) of model i relative to the base model is $\Delta_i^l = \theta_i^l - \theta_{\text{pre}}^l$, where $\Delta_i^l \in \mathbb{R}^{m \times d}$ is the weight difference for a layer of dimensions $m \times d$.

3.1 LOW-RANK COMPRESSED MODELS

In practice, fine-tuned models are frequently compressed into low-rank form to reduce storage, training, and inference costs. We focus on two widely used strategies:

LoRA fine-tuning: During training, each layer is augmented with trainable low-rank matrices $A \in \mathbb{R}^{m \times r}$ and $B \in \mathbb{R}^{r \times d}$, where $r \ll \min(m, d)$. The pretrained weights remain frozen, so adaptation is concentrated entirely in A and B .

Post-training SVD truncation (PT-SVD): After standard fine-tuning, the update Δ_i^l is decomposed with SVD and truncated to the top- r singular components. This yields the low-rank approximation $\hat{\Delta}_i^l = A_i^l B_i^l$, for $A_i^l \in \mathbb{R}^{m \times r}$ and $B_i^l \in \mathbb{R}^{r \times d}$ where $r \ll \min(m, d)$.

In both cases, the fine-tuned update for layer l in model i is stored as $\hat{\Delta}_i^l = A_i^l B_i^l$, where $A_i^l \in \mathbb{R}^{m \times r}$ and $B_i^l \in \mathbb{R}^{r \times d}$, with r denoting the compression rank. The merging problem therefore becomes the problem of combining low-rank updates across tasks. Under the conventional single-model objective, the merged model integrates the compressed deltas $\hat{\Delta}_i = \{\hat{\Delta}_i^l\}_{l=1}^L$ across all tasks:

$$\hat{\theta}_{\mathcal{M}} = \theta_{\text{pre}} + \mathcal{M}(\{\hat{\Delta}_i\}_{i=1}^n), \quad \max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\hat{\theta}_{\mathcal{M}}, \mathcal{D}_i), \quad (2)$$

where L is the total number of layers. This paper focuses on exactly this low-rank setting. The next section shows that standard merging strategies are not merely suboptimal here; they are structurally mismatched to compressed representations.

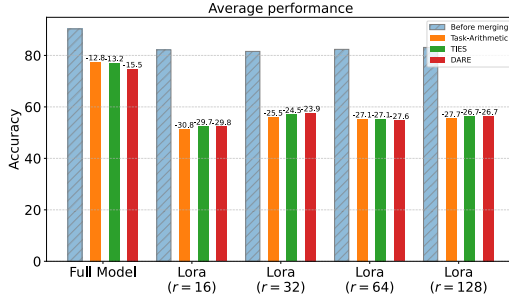


Figure 1: Average task performance of individual models and merged models under different merging methods. The comparison shows that LoRA-compressed models at ranks $r \in \{16, 32, 64, 128\}$ suffer sharp degradation under conventional merging relative to the unmerged task-specific models.

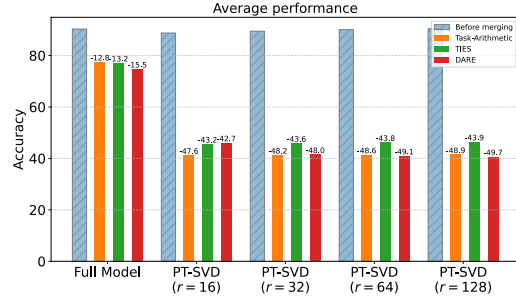


Figure 2: Average task performance of individual models and merged models under different merging methods. The comparison shows that PT-SVD-compressed models at ranks $r \in \{16, 32, 64, 128\}$ expose the same failure mode: standard merging substantially underperforms the unmerged task-specific models.

4 MOTIVATING OBSERVATION

We now test conventional model merging directly in the low-rank setting introduced above. For each layer l , prior work considers two natural extensions of merging to compressed weights (Tang et al., 2025; Zhao et al., 2025; Zheng et al., 2025; Zhang et al., 2025):

- **Combined merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{\mathbf{A}_i^l \mathbf{B}_i^l\}_{i=1}^n),$$

where each low-rank update is first reconstructed into full-rank form and then merged.

- **Separate merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{\mathbf{A}_i^l\}_{i=1}^n) \mathcal{M}(\{\mathbf{B}_i^l\}_{i=1}^n),$$

where the low-rank factors are merged independently and then recomposed.

Both formulations are mathematically valid, but only the separate strategy preserves the practical value of compression. The combined strategy stores the merged update as an $m \times d$ matrix, whereas the separate strategy stores the low-rank factors $\mathbf{A}_i^l \in \mathbb{R}^{m \times r}$ and $\mathbf{B}_i^l \in \mathbb{R}^{r \times d}$. Since $r(m+d) \ll md$ for $r \ll \min(m, d)$, the combined approach reintroduces the very storage and memory costs that compression is meant to remove. For a layer with $m = d = 1024$ and rank $r = 16$, the combined approach stores $md = 1,048,576$ parameters, while the separate approach stores only $r(m+d) = 16 \times (1024 + 1024) = 32,768$ parameters.

This is a **32 $\times$ reduction in storage**, making clear that combined merging defeats the purpose of low-rank compression. We therefore focus on the separate strategy as the practical baseline for merging compressed models.

Using the separate strategy, we applied several established merging baselines to low-rank compressed models across multiple compression ratios. We use RoBERTa-base (Zhuang et al., 2021) as the pre-trained model and evaluate on 5 GLUE tasks (Wang et al., 2018) (details in Appendix A.3). Figures 1 and 2 show a clear result: merging compressed models produces a *catastrophic drop* in average task performance compared with the corresponding full-rank setting. We identify two concrete causes: (i) **Amplified task interference**: compression discards much of the representation capacity, leaving low-rank models more susceptible to destructive interference when merged; (ii) **Loss of coupling between $\mathbf{A}$ and $\mathbf{B}$** : conventional merging treats low-rank matrices $\mathbf{A}_i^l$ and $\mathbf{B}_i^l$ as independent factors, ignoring the coupled nature of their product, leading to structural misalignment in the merged updates. Together, these mechanisms severely impair merged low-rank models and reveal a deeper incompatibility between traditional single-model merging and compressed representations.

This evidence leads to the core insight of the paper: under conventional strategies, a single merged model cannot reliably recover the task-specific performance of the original low-rank adapters. The

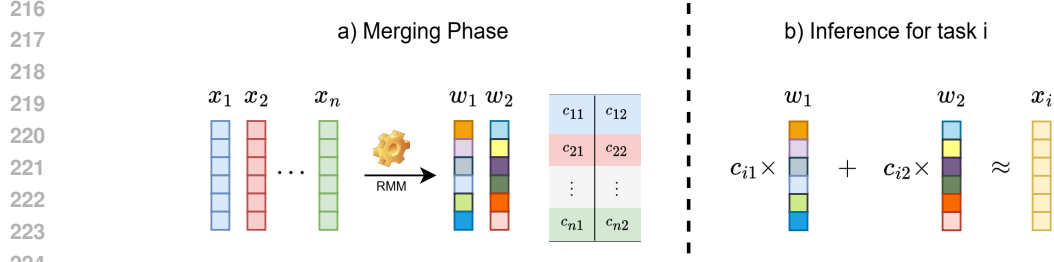


Figure 3: Overview of the RMM framework. **Left:** The merging phase computes a compact set of basis directions (two in this example) and the associated task-specific coefficients. **Right:** At inference, RMM reconstructs target model i by linearly combining the basis vectors with the corresponding coefficients.

better objective is to reconstruct the original compressed models, or close approximations of them, from a compact set of basis models whose size reflects the storage budget. Rather than forcing all updates into one representation, we define merging as the construction of a *reconstruction-capable model space*. We now formalize this objective and introduce *Reversible Model Merging* (RMM).

5 REVERSIBLE MODEL MERGING

The empirical failures above establish a fundamental limitation of compressed-model merging: one merged model cannot faithfully preserve the task-specific performance of all low-rank compressed models. RMM addresses this limitation by changing the target representation. It does not force all models into a single weight set; instead, it constructs a small set of shared components (basis models) from which the original task models can be approximately reconstructed.

Instead of compressing all n task-specific models into one merged solution $\hat{\theta}_{\mathcal{M}}$, RMM retains p merged components, where $2 \leq p \ll n$. These components define a low-dimensional subspace that captures the essential variation across tasks and supports efficient recovery of each model on demand. The design makes the performance–memory trade-off explicit: increasing p improves reconstruction capacity, while decreasing p reduces storage.

For each layer l , RMM operates on the r -dimensional vectors inside the low-rank delta factors (A_i^l and B_i^l): rows of A_i^l and columns of B_i^l , where r is the compression rank. We treat each row of A_i^l and each column of B_i^l as an independent *task vector*, and we call a fixed row or column index a *task vector position*. For any fixed position, we collect the corresponding task vectors across all n models and denote the vector from model i as $x_i \in \mathbb{R}^r$. RMM is applied independently at each position: it compresses $\{x_i\}_{i=1}^n$ into a shared basis and later reconstructs the required vector from that basis. Each layer has $m + d$ such positions, m rows from A_i^l and d columns from B_i^l , and the same procedure is applied separately to each position.

For each position, we learn basis vectors $\{w_1, \dots, w_p\}$, with $w_j \in \mathbb{R}^r$ for $1 \leq j \leq p$. Their span defines $W = [w_1 \dots w_p] \in \mathbb{R}^{r \times p}$, the subspace used to reconstruct each task vector x_i . We also learn a coefficient matrix $C \in \mathbb{R}^{n \times p}$, whose i -th row $[c_{i1}, \dots, c_{ip}]$ gives the scalar weights for reconstructing x_i from the basis. The optimization problem is therefore exact in intent: choose W and C to minimize total reconstruction error across all n task vectors:

$$\min_{W, C} \sum_{i=1}^n \left\| x_i - \sum_{j=1}^p c_{ij} w_j \right\|_2^2. \quad (3)$$

This representation stores rp basis parameters and np coefficient parameters, for a total of $p(r + n)$ instead of the rn parameters needed to store all task vectors directly. The hyperparameter p directly controls the storage–performance trade-off. Figure 3 illustrates RMM for $p = 2$. The optimization in equation 3 has the following matrix form:

$$\min_{W, C} \|X - CW^\top\|_F^2, \quad (4)$$

where $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$ contains all task vectors. For any fixed $\mathbf{W}$, the optimal coefficients are given by the least-squares solution $\mathbf{C}^* = \mathbf{X}\mathbf{W}(\mathbf{W}^\top \mathbf{W})^{-1}$. With orthonormal basis vectors ($\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p, p < r$), the problem reduces to:

$$\min_{\mathbf{W}} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 \quad \text{s.t.} \quad \mathbf{W}^\top \mathbf{W} = \mathbf{I}_p. \quad (5)$$

Thus, RMM stores $\mathbf{W} \in \mathbb{R}^{r \times p}$ and $\mathbf{X}\mathbf{W} \in \mathbb{R}^{n \times p}$, where $\mathbf{X}\mathbf{W}$ is the coefficient matrix $\mathbf{C}$ in equation 4.

Theorem 1 (Optimal Solution to equation 5). *Let $\mathbf{X} \in \mathbb{R}^{n \times r}$ be zero-centered and $p < \min\{r, n\}$. The optimal basis $\mathbf{W}^*$ that minimizes equation 5 is given by the top- p eigenvectors of the sample covariance matrix $\mathbf{X}^\top \mathbf{X}$. Equivalently, $\mathbf{W}^*$ consists of the top- p right singular vectors of $\mathbf{X}$.*

The proof of Theorem 1 is provided in Appendix A.1. This result gives RMM a closed-form optimal basis for the reconstruction objective. We now describe the two phases of the method.

5.1 MERGING PHASE

The merging phase constructs the compact representation used by all task-specific models. For each layer and task vector position (a fixed row in $\mathbf{A}_i^l$ or column in $\mathbf{B}_i^l$), we gather task vectors across the n models as $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$. We compute the mean vector $\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ and zero-center $\mathbf{X}$ (i.e., $\mathbf{X} \leftarrow [\mathbf{x}_1 - \boldsymbol{\mu} \dots \mathbf{x}_n - \boldsymbol{\mu}]^\top$). We then perform SVD on the centered matrix $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ and select the top- p right singular vectors, yielding the optimal orthonormal basis $\mathbf{W}^*$ for equation 5: $\mathbf{W}^* = \mathbf{V}_{[:,1:p]} \in \mathbb{R}^{r \times p}$. Finally, we compute the coefficient matrix $\mathbf{C}^* = \mathbf{X}\mathbf{W}^* \in \mathbb{R}^{n \times p}$, whose i -th row $[c_{i1}, \dots, c_{ip}]$ stores the weights needed to reconstruct $\mathbf{x}_i$.

After this phase, the original task vectors $\{\mathbf{x}_i\}_{i=1}^n$ can be discarded. RMM retains only $\mathbf{W}^*, \mathbf{C}^*$, and $\boldsymbol{\mu}$ for each task vector position, reducing storage from rn parameters to $p(r+n) + r$ parameters per position. The entire merging phase is performed offline before inference.

5.2 INFERENCE PHASE

Following standard model-merging practice, we assume the target task for each input is either specified by the user or selected by an oracle router that assigns the input to the appropriate model index. Once the target index i is known, RMM reconstructs its task vector $\mathbf{x}_i$, denoted $\hat{\mathbf{x}}_i$, from the compact representation $(\mathbf{W}^*, \mathbf{C}^*, \boldsymbol{\mu})$ stored for each task vector position. It then recovers $\mathbf{A}_i^l$ and $\mathbf{B}_i^l$ by concatenating the reconstructed vectors across all positions, i.e., the corresponding rows of $\mathbf{A}_i^l$ and columns of $\mathbf{B}_i^l$.

For each task vector position, $\mathbf{W}^* \in \mathbb{R}^{r \times p}$ contains the orthonormal basis vectors, $\mathbf{C}^* \in \mathbb{R}^{n \times p}$ stores task-specific coefficients, and $\boldsymbol{\mu} \in \mathbb{R}^r$ is the mean vector across task vectors. RMM reconstructs the target vector by linearly combining the basis directions with the task-specific coefficients and shifting by $\boldsymbol{\mu}$:

$$\hat{\mathbf{x}}_i = \sum_{j=1}^p c_{ij}^* \mathbf{w}_j^* + \boldsymbol{\mu} = \mathbf{C}_{[i,:]}^* \mathbf{W}^{*\top} + \boldsymbol{\mu}.$$

The reconstructed vector $\hat{\mathbf{x}}_i \in \mathbb{R}^r$ approximates the original task vector $\mathbf{x}_i$ discarded during merging. Applying this reconstruction independently to every task vector position and every layer recovers the full low-rank delta for task-specific model i , denoted $\hat{\Delta}_i$. Algorithm 1 summarizes the complete merging and inference procedure. We next evaluate how this reversible construction performs against standard baselines.

6 EXPERIMENTS

We evaluate RMM across compression settings, tasks, and model architectures to test the central claim of the paper: reversible merging preserves the performance of low-rank task-specific models far better than conventional single-model merging while still reducing storage.

Algorithm 1 Reversible Model Merging (RMM)

```
1: Merging Phase:
2: Require: Low-rank deltas  $A_i^l$  and  $B_i^l$ , compression rank  $r$ , number of basis vectors  $p$ 
3: for  $l = 1$  to  $L$  do
4:   for each task vector position (each row in  $A_i^l$  or each column in  $B_i^l$ ) do
5:     Gather task vectors  $x_i$  of each model  $i$  ( $1 \leq i \leq n$ )
6:      $X \leftarrow [x_1 \dots x_n]^\top \in \mathbb{R}^{n \times r}$ 
7:      $\mu \leftarrow \sum_{i=1}^n x_i \in \mathbb{R}^r$ 
8:      $X \leftarrow [x_1 - \mu \dots x_n - \mu]^\top$ 
9:      $U, \Sigma, V^\top \leftarrow \text{SVD}(X)$ 
10:     $W^* \leftarrow V_{[:,1:p]} \in \mathbb{R}^{r \times p}$ 
11:     $C^* \leftarrow XW^* \in \mathbb{R}^{n \times p}$ 
12:    Store  $(W^*, C^*, \mu)$  and discard task vectors  $\{x_1 \dots x_n\}$ 
13:   end for
14: end for
15: Inference Phase:
16: Require: target task  $i$ , stored representation  $(W^*, C^*, \mu)$  for each task vector position
17: for  $l = 1$  to  $L$  do
18:   for each task vector position do
19:      $\hat{x}_i = C_{[i,:]}^* W^{*\top} + \mu$ 
20:   end for
21:    $\hat{A}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all row positions})$ 
22:    $\hat{B}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all column positions})$ 
23: end for
24:  $\hat{\Delta}_i = \{\hat{A}_i^l \hat{B}_i^l\}_{l=1}^L$ 
25: Inference with the reconstructed model  $\theta_{pre} + \hat{\Delta}_i$ 
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6.1 EXPERIMENTAL SETUP

We first evaluate RMM on eight RoBERTa-base models fine-tuned for eight distinct tasks (Zhuang et al., 2021). Each model is compressed independently with either PT-SVD or LoRA at ranks $r \in \{16, 32, 64, 128\}$. We then apply RMM to the resulting low-rank models and compare against standard merging baselines: Task Arithmetic (TA) (Ilharco et al., 2023), TIES-merging (Yadav et al., 2023), and DARE (Yu et al., 2024). We also report the individual compressed models before merging as the reference upper bound. Appendix A.2 extends the evaluation to OPT-1.3b (Zhang et al., 2022) and ViT-B/32 (Radford et al., 2021), covering additional model scales, modalities, and task counts.

Following the GLUE benchmark protocol (Wang et al., 2018), we use the standard metric for each task. QNLI (question–answering inference), MRPC (paraphrase detection), SST-2 (sentiment classification), MNLI (multi-genre natural language inference), QQP (duplicate question detection), and RTE (textual entailment) are measured by accuracy. CoLA (linguistic acceptability) uses Matthews correlation, and STS-B (semantic textual similarity) uses Pearson correlation. We report the average across tasks as the overall performance measure. We also report relative storage, measured as the fraction of parameters retained compared with storing all task-specific adapters. Single-model baselines have fixed cost $\frac{r}{rn} = \frac{1}{n}$. For RMM with p basis components, the storage cost is $\frac{p(r+n)+r}{rn}$, as derived in Section 5.1.

6.2 RESULTS AND DISCUSSION

Tables 1 and 2 report the main comparison under PT-SVD and LoRA compression. Across both compression regimes and all ranks $r \in \{16, 32, 64, 128\}$, RMM decisively outperforms TA, TIES, and DARE. At low ranks ($r = 16, 32$), the single-model baselines collapse to only 31–42% average score and fail especially badly on structurally sensitive tasks such as CoLA and STS-B. RMM reverses this degradation: with $p = 2$, it achieves substantial gains (e.g., 57.24% under PT-SVD and 51.41% under LoRA at $r = 16$), and with $p = 3$, it reaches stronger averages (up to 72.22% and

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	77.98	58.05	90.59	83.80
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.19	31.66
	TIES	13%	50.56	34.78	49.08	31.82	36.82	47.29	0.00	3.75	31.76
	DARE	13%	50.80	33.51	49.08	31.82	36.82	47.29	0.00	5.07	31.80
	RMM ($p = 2$)	50%	51.20	78.20	88.30	66.74	86.02	47.29	6.56	33.59	57.24
	RMM ($p = 3$)	69%	88.83	86.26	92.89	79.58	87.18	48.74	33.85	60.46	72.22
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	78.34	59.33	90.68	84.46
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.21	31.66
	TIES	13%	50.56	35.19	49.08	31.82	36.82	47.29	0.00	3.87	31.83
	DARE	13%	50.72	33.51	49.08	31.82	36.82	47.29	0.00	3.48	31.59
	RMM ($p = 2$)	44%	50.54	74.32	55.05	62.68	87.07	47.29	0.00	23.18	50.02
	RMM ($p = 3$)	59%	84.66	86.32	92.89	79.70	88.05	47.29	23.57	44.54	68.38
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	78.70	61.11	90.72	85.10
	TA	13%	50.61	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.13	49.08	31.82	36.82	47.29	0.00	3.80	31.82
	DARE	13%	50.58	33.51	49.08	31.82	36.82	47.29	0.00	5.22	31.79
	RMM ($p = 2$)	41%	50.53	73.10	49.08	58.48	86.98	47.29	0.00	17.50	47.87
	RMM ($p = 3$)	55%	74.14	86.20	92.43	78.83	89.00	47.29	13.16	33.70	64.34
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	79.78	60.16	90.74	85.30
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.54	49.08	31.82	36.82	47.29	0.00	3.93	31.88
	DARE	13%	50.61	34.78	49.08	31.82	36.82	47.29	0.00	2.41	31.60
	RMM ($p = 2$)	39%	50.54	70.78	49.08	54.69	84.68	47.29	0.00	14.53	46.45
	RMM ($p = 3$)	52%	59.20	85.80	92.32	77.50	89.02	47.29	6.56	26.94	60.58

Table 1: GLUE performance for merging eight RoBERTa-base models compressed with PT-SVD at ranks $r = 16, 32, 64, 128$. RMM retains substantially more task performance than single-model merging baselines across all ranks.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	68.23	53.92	88.71	77.71
	TA	13%	50.59	33.51	72.13	34.79	64.46	47.29	0.00	21.81	40.57
	TIES	13%	50.63	33.68	74.31	35.64	67.45	47.29	0.00	26.38	41.92
	DARE	13%	50.63	33.51	73.74	35.97	65.35	47.29	0.00	21.62	41.01
	RMM ($p = 2$)	50%	51.13	34.14	90.71	60.48	84.41	47.29	0.00	43.14	51.41
	RMM ($p = 3$)	69%	55.92	38.14	93.12	82.95	87.30	47.29	6.56	57.81	58.64
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	62.82	53.12	88.53	76.54
	TA	13%	50.91	68.06	54.70	33.77	65.50	55.96	0.00	-0.48	41.05
	TIES	13%	51.03	68.06	56.77	34.24	66.27	58.12	0.00	1.02	41.94
	DARE	13%	50.96	64.64	63.07	35.12	63.66	55.60	0.00	1.01	41.76
	RMM ($p = 2$)	44%	52.08	65.39	81.54	62.46	77.06	57.40	0.00	9.95	50.73
	RMM ($p = 3$)	59%	54.62	69.97	92.20	81.29	85.97	57.40	4.64	24.08	58.77
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	67.15	58.31	88.70	78.25
	TA	13%	54.75	68.52	77.29	37.24	36.82	47.29	0.00	-3.53	39.80
	TIES	13%	54.82	68.75	80.73	37.67	36.82	47.29	0.00	-2.48	40.45
	DARE	13%	53.71	68.58	78.56	38.33	36.82	47.29	0.00	-4.28	39.88
	RMM ($p = 2$)	41%	55.74	69.28	87.27	42.81	77.72	47.29	0.00	4.91	48.13
	RMM ($p = 3$)	55%	59.07	70.14	90.83	79.11	82.06	47.29	0.00	14.70	55.40
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	57.04	59.35	89.68	77.71
	TA	13%	51.49	69.22	54.47	37.16	63.27	47.29	0.00	-1.74	40.15
	TIES	13%	51.91	68.93	57.34	37.81	63.36	47.29	0.00	-0.25	40.80
	DARE	13%	51.97	68.99	58.60	35.64	63.33	47.29	0.00	-0.44	40.67
	RMM ($p = 2$)	39%	57.18	67.83	88.53	49.54	84.64	47.29	0.00	28.00	52.88
	RMM ($p = 3$)	52%	59.13	39.19	92.09	80.08	86.39	47.29	15.45	71.18	61.35

Table 2: GLUE performance for merging eight RoBERTa-base models compressed with LoRA at ranks $r = 16, 32, 64, 128$. RMM consistently improves over single-model merging baselines while using a controllable amount of additional offline storage.

58.64% at $r = 16$), sharply narrowing the gap to the individual compressed models. At higher ranks ($r = 64, 128$), where compression error is smaller, RMM still dominates. With $p = 3$, it achieves 64.34% and 60.58% under PT-SVD, and 55.40% and 61.35% under LoRA, while the single-model baselines remain near 40%. Even $p = 2$ preserves a clear advantage, confirming that reversible reconstruction directly mitigates the destructive interference that limits prior methods.

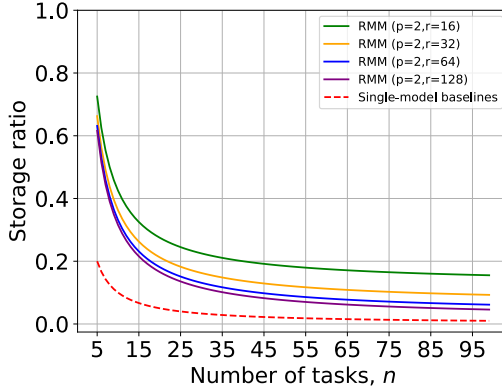


Figure 4: Scalability of RMM ($p = 2$). The storage ratio reports memory relative to storing all models and decreases as the number of tasks grows.

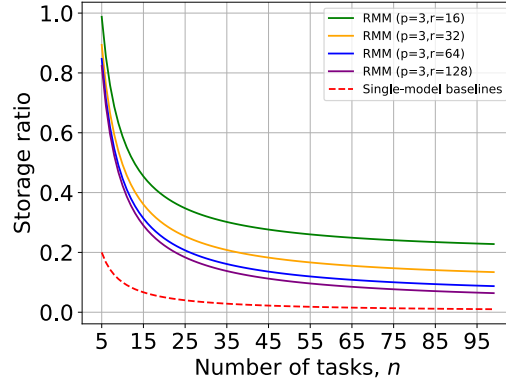


Figure 5: Scalability of RMM ($p = 3$). The storage ratio reports memory relative to storing all models and confirms sublinear growth with task count.

The relative gain of RMM over baselines decreases as rank increases, which is expected because higher-rank task representations occupy richer and less aligned spaces that are harder to summarize with a fixed low-dimensional basis. Nevertheless, the overall conclusion is stable: RMM provides a consistent advantage across compression regimes and ranks. It preserves high task performance by spending additional storage in a controlled way, and the number of reversible components directly tunes the accuracy–storage trade-off. Crucially, this storage is **offline memory**, typically CPU-side and inexpensive, whereas task performance determines runtime quality. The added storage is therefore a practical and favorable exchange for the observed accuracy gains.

6.3 SCALABILITY OF RMM

We next examine scalability as the number of models, n , increases. Figures 4 and 5 plot the storage ratio relative to storing all individual task-specific models for RMM with $p = 2$ and $p = 3$. The result is favorable and direct: as the number of tasks grows, the relative storage cost decreases substantially. This trend is a defining benefit of reversible merging. RMM storage grows sublinearly with the number of task-specific models because, once the basis is fixed, each additional task requires only an additional row in the coefficient matrix. Traditional choices are less satisfactory: a single merged model sacrifices performance, while storing every task-specific model scales linearly. RMM therefore provides a practical route to large-scale multi-task, continual, and federated deployments where many models must be managed efficiently without giving up task performance.

7 CONCLUSION

We introduced a principled framework for reversible model merging that changes how multiple fine-tuned models are stored, merged, and reconstructed. Instead of forcing all task-specific knowledge into one merged model, RMM stores a compact set of intermediate representations—far fewer than the original models, but far more expressive than a single aggregate. This design reconstructs task-specific models accurately while substantially reducing storage overhead. It provides a controllable performance–storage trade-off for low-rank compressed models and demonstrates that compact bases with reversible mappings can support high-accuracy deployment under strict memory budgets. Practitioners can therefore choose the degree of compression and reconstruction fidelity required by their application.

Overall, RMM moves beyond heuristic and one-shot merging by formalizing merging as a structured, reversible process. This perspective broadens the design space for efficient multi-task deployment and establishes a clear direction for future work on performance-aware compression and reconstruction strategies.

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A APPENDIX

A.1 PROOF OF THEOREM 1

Let $\mathbf{A} = \mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top$. The Frobenius norm identity gives $\|\mathbf{A}\|_F^2 = \text{Tr}(\mathbf{A}^\top \mathbf{A})$, where $\text{Tr}(\cdot)$ denotes the trace of a square matrix. Expanding the reconstruction objective yields:

$$\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 = \text{Tr}((\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)).$$

The term inside the trace expands as:

$$\begin{aligned} (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) &= (\mathbf{X}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top)(\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) \\ &= \mathbf{X}^\top \mathbf{X} - \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X} \\ &\quad + \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top. \end{aligned}$$

Taking the trace of this expression and using cyclicity together with $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p$ gives:

$$\begin{aligned} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}). \end{aligned}$$

Since $\text{Tr}(\mathbf{X}^\top \mathbf{X})$ is constant with respect to $\mathbf{W}$, minimizing $\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2$ is exactly equivalent to maximizing $\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W})$. Thus,

$$\min_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 \iff \max_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}).$$

The Rayleigh quotient theorem (Horn & Johnson, 2012) then identifies the maximizer: the columns of $\mathbf{W}$ must be the top- p eigenvectors of $\mathbf{X}^\top \mathbf{X}$. This is the principal component analysis (PCA) solution, which selects the p -dimensional subspace that maximizes projected variance and, equivalently, minimizes squared reconstruction error. If $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ is the SVD of $\mathbf{X}$, then $\mathbf{X}^\top \mathbf{X} = \mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^\top$, so the eigenvectors of $\mathbf{X}^\top \mathbf{X}$ are exactly the right singular vectors (columns of $\mathbf{V}$). Therefore, RMM obtains the same optimal basis by selecting the top- p right singular vectors of $\mathbf{X}$. In practice, this SVD route is numerically stable and often preferable to explicitly forming $\mathbf{X}^\top \mathbf{X}$, especially when r is large.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	BOOLQ	Average
16	No merging	100%	91.74	71.54	95.30	84.19	88.14	71.47	83.73
	Task-Arithmetic	17%	50.34	37.16	58.49	34.43	63.30	41.87	47.60
	TIES	17%	49.79	36.75	61.24	35.57	63.55	40.67	47.93
	DARE	17%	49.95	37.68	56.08	34.51	63.39	42.78	47.40
	RMM ($p = 2$)	63%	88.12	44.00	93.58	73.06	85.63	59.69	74.01
	RMM ($p = 3$)	85%	91.58	49.04	95.64	80.31	87.52	65.50	78.27
32	No merging	100%	92.06	72.17	95.41	84.60	89.17	72.45	84.31
	Task-Arithmetic	17%	50.28	37.22	58.94	34.57	63.33	41.93	47.71
	TIES	17%	49.77	36.93	61.93	35.69	63.65	40.73	48.12
	DARE	17%	50.39	40.87	56.19	34.96	63.59	43.33	48.22
	RMM ($p = 2$)	56%	86.66	41.91	91.74	71.75	85.68	56.39	72.36
	RMM ($p = 3$)	76%	91.85	44.81	94.84	80.21	88.43	63.52	77.28
64	No merging	100%	92.26	72.12	95.41	84.64	90.07	72.84	84.56
	Task-Arithmetic	17%	50.28	37.28	59.06	34.56	63.34	42.08	47.77
	TIES	17%	49.75	37.04	61.81	35.84	63.74	40.76	48.16
	DARE	17%	50.16	40.52	58.83	35.59	63.35	42.54	48.50
	RMM ($p = 2$)	53%	83.87	42.72	89.45	69.76	85.70	53.43	70.82
	RMM ($p = 3$)	71%	92.07	41.97	94.50	79.70	88.73	61.96	76.49
128	No merging	100%	92.35	72.41	95.41	84.67	90.93	73.09	84.81
	Task-Arithmetic	17%	50.32	37.39	58.94	34.63	63.37	42.08	47.79
	TIES	17%	49.79	37.04	62.04	35.99	63.74	40.55	48.19
	DARE	17%	50.03	38.14	57.91	34.72	63.31	42.11	47.70
	RMM ($p = 2$)	52%	81.38	42.78	84.75	67.66	83.22	51.99	68.63
	RMM ($p = 3$)	69%	92.28	41.80	93.69	79.42	88.76	59.69	75.94

Table 3: GLUE performance for merging six OPT-1.3b models compressed with PT-SVD at ranks $r = 16, 32, 64, 128$. RMM consistently improves over the single-model merging baselines while preserving much of the individual-model performance.

r	Method	Storage	MNIST	GTSRB	SVHN	Cars	EuroSAT	DTD	SUN397	Average
16	No merging	100%	99.61	97.22	97.06	69.20	99.67	75.21	72.72	87.24
	Task-Arithmetic	14%	47.19	28.29	38.64	56.27	57.52	41.38	62.92	47.46
	TIES	14%	60.14	34.52	49.03	58.55	61.70	43.67	62.87	52.93
	DARE	14%	53.80	31.00	44.96	57.60	56.93	43.99	62.42	50.10
	RMM ($p = 2$)	55%	99.16	88.52	94.89	66.61	97.44	60.64	67.86	82.16
	RMM ($p = 3$)	76%	99.58	95.65	96.54	68.30	98.93	67.18	69.50	85.10
32	No merging	100%	99.65	98.45	97.35	73.04	99.78	78.72	75.52	88.93
	Task-Arithmetic	14%	47.90	28.31	38.68	56.24	57.52	41.38	63.00	47.58
	TIES	14%	61.25	35.28	49.95	58.80	62.41	44.04	63.05	53.54
	DARE	14%	53.05	31.04	46.34	58.54	57.74	42.87	62.87	50.35
	RMM ($p = 2$)	49%	98.98	88.23	94.29	68.32	96.81	61.91	68.36	82.41
	RMM ($p = 3$)	67%	99.50	96.53	96.68	70.90	98.93	70.43	71.23	86.31
64	No merging	100%	99.67	98.68	97.41	75.76	99.85	79.15	77.73	89.75
	Task-Arithmetic	14%	47.99	28.30	38.71	56.21	57.59	41.38	63.22	47.63
	TIES	14%	61.83	35.93	50.28	59.08	63.07	44.15	63.17	53.93
	DARE	14%	54.78	32.79	45.05	58.23	61.30	42.87	62.77	51.11
	RMM ($p = 2$)	46%	98.46	84.43	92.83	69.36	95.59	62.39	69.77	81.83
	RMM ($p = 3$)	62%	99.45	96.14	96.33	72.84	98.56	72.07	73.85	87.03
128	No merging	100%	99.68	98.75	97.46	76.99	99.85	79.04	78.59	90.05
	Task-Arithmetic	14%	48.00	28.31	38.73	56.21	57.63	41.44	63.35	47.67
	TIES	14%	62.08	36.36	50.48	59.23	63.41	44.68	63.25	54.21
	DARE	14%	55.16	32.07	45.47	58.55	61.85	43.09	62.85	51.29
	RMM ($p = 2$)	44%	97.22	78.56	90.63	70.22	94.00	61.49	71.03	80.45
	RMM ($p = 3$)	59%	99.28	94.59	95.95	74.24	98.19	71.86	76.02	87.16

Table 4: Vision-task performance for merging seven ViT-B/32 models compressed with PT-SVD at ranks $r = 16, 32, 64, 128$. RMM delivers strong reconstruction quality across datasets and scales favorably in storage.

A.2 ADDITIONAL RESULTS

This section extends the evidence for RMM to two additional pre-trained models: OPT-1.3b and ViT-B/32. For **OPT-1.3b**, we evaluate six natural language understanding tasks from GLUE and related benchmarks: QNLI, MRPC, SST-2, MNLI, QQP, and BoolQ (Clark et al., 2019). For **ViT-B/32**, we evaluate seven diverse vision classification datasets: MNIST (LeCun et al., 1998), GTSRB

(Stallkamp et al., 2011), SVHN (Netzer et al., 2011), Cars (Krause et al., 2013), EuroSAT (Helber et al., 2019), DTD (Cimpoi et al., 2014), and SUN397 (Xiao et al., 2010).

Table 3 reports results for six OPT-1.3b models compressed with PT-SVD. Both RMM variants ($p = 2$ and $p = 3$) consistently outperform the baselines, and $p = 3$ closes much of the gap to the individual models while requiring substantially fewer parameters. Table 4 reports results for seven ViT-B/32 models. The vision results are even stronger: with $p = 3$, RMM achieves average accuracy less than 3% below the original models while using only 59% of the storage required to retain all models at $r = 128$. These results reinforce the main conclusion that RMM reconstructs compressed models faithfully while delivering meaningful storage savings.

A.3 RESULTS FOR 5 TASKS

For the motivating experiments in Section 4, we use RoBERTa-base as the pretrained model and evaluate merging baselines on five GLUE tasks: QNLI, MRPC, SST-2, MNLI, and QQP. Figures 1 and 2 in Section 4 summarize the main failure pattern. Table 5 provides the LoRA results corresponding to Figure 1, including the improvements from RMM, and Table 6 provides the PT-SVD results corresponding to Figure 2.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	82.16
	Task-Arithmetic	20%	50.61	33.51	73.85	34.23	64.58	51.36
	TIES	20%	50.65	33.51	75.57	35.83	66.97	52.51
	DARE	20%	50.61	33.51	79.36	34.70	63.40	52.32
	RMM ($p = 2$)	73%	59.64	68.64	93.00	82.68	86.89	78.17
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	81.57
	Task-Arithmetic	20%	51.18	67.30	58.83	34.33	68.56	56.04
	TIES	20%	51.36	68.17	59.52	34.76	71.46	57.05
	DARE	20%	50.83	64.87	68.12	35.54	68.93	57.66
	RMM ($p = 2$)	66%	57.72	67.54	92.55	81.80	86.13	77.15
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	82.37
	Task-Arithmetic	20%	54.74	68.93	80.28	35.70	36.82	55.29
	TIES	20%	55.26	68.93	78.78	36.76	36.82	55.31
	DARE	20%	54.71	68.81	74.54	38.72	36.82	54.72
	RMM ($p = 2$)	63%	59.82	67.48	91.51	83.21	84.41	77.29
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	83.12
	Task-Arithmetic	20%	52.52	68.70	54.82	37.83	63.46	55.47
	TIES	20%	52.55	69.04	57.80	38.91	63.72	56.40
	DARE	20%	55.04	69.62	54.47	38.92	64.12	56.43
	RMM ($p = 2$)	62%	60.21	34.96	91.86	78.59	86.65	70.45

Table 5: GLUE performance for merging five RoBERTa-base models compressed with LoRA at ranks $r = 16, 32, 64, 128$. RMM substantially improves over conventional low-rank merging in the motivating setting.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	88.76
	Task-Arithmetic	20%	53.03	35.19	49.08	31.82	36.82	41.19
	TIES	20%	55.46	54.32	49.08	31.82	37.21	45.58
	DARE	20%	53.69	58.67	49.08	31.82	37.08	46.07
	RMM ($p = 2$)	73%	79.72	85.16	90.83	76.58	86.09	83.68
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	89.46
	Task-Arithmetic	20%	53.23	35.59	49.08	31.82	36.82	41.31
	TIES	20%	55.78	55.19	49.08	31.82	37.62	45.90
	DARE	20%	55.89	33.74	49.20	31.82	36.85	41.50
	RMM ($p = 2$)	66%	63.87	82.78	91.28	77.12	87.05	80.42
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	90.05
	Task-Arithmetic	20%	53.49	35.88	49.08	31.82	36.82	41.42
	TIES	20%	56.14	55.94	49.08	31.82	38.08	46.21
	DARE	20%	53.51	33.68	49.08	31.82	36.82	40.98
	RMM ($p = 2$)	63%	56.12	81.22	91.06	76.74	88.07	78.64
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	90.34
	Task-Arithmetic	20%	53.60	36.06	49.08	31.82	36.82	41.48
	TIES	20%	56.40	56.35	49.08	31.82	38.61	46.45
	DARE	20%	50.52	34.96	49.08	31.82	36.92	40.66
	RMM ($p = 2$)	62%	54.11	79.19	90.60	74.49	87.46	77.17

Table 6: GLUE performance for merging five RoBERTa-base models compressed with PT-SVD at ranks $r = 16, 32, 64, 128$. RMM preserves far more task performance than the single-model baselines.